

**Predicting Median House Values in Philadelphia:
An OLS Regression Analysis**

By Darren Marcus Loo

1. Introduction

Residential property values are shaped not only by the physical attributes of individual housing units, but also by the broader socioeconomic characteristics of the neighbourhoods in which they are situated. Understanding what drives spatial variation in housing prices is of considerable practical importance for urban planners, policymakers, and researchers alike, as housing wealth represents a primary component of household financial assets and a key indicator of neighbourhood well-being. This study examines the relationship between median house values and several neighbourhood-level characteristics in Philadelphia, Pennsylvania, using data at the Census block group level. Philadelphia presents a particularly compelling setting for this analysis, given the stark spatial disparities in socioeconomic conditions documented across its metropolitan area (Gyourko and Summers, 2007). Using Ordinary Least Squares (OLS) regression, we model median house values as a function of educational attainment, poverty, vacancy rates, and the prevalence of single-family housing across 1,720 block groups.

A substantial body of literature supports the use of neighbourhood socioeconomic characteristics as predictors of residential property values. Hedonic pricing theory posits that house prices reflect a bundle of attributes (structural, locational, and neighbourhood-level), and that buyers implicitly pay for each characteristic (Li and Brown, 1980). Within this framework, neighbourhood composition variables have been shown to exert significant effects on house values. Dubin and Goodman (1982) demonstrate that educational attainment and poverty levels are meaningful determinants of housing prices in inner-city areas, with better-educated and less impoverished neighbourhoods commanding higher values. This is consistent with the expectation that areas with higher proportions of bachelor's degree holders (PCBACHMORE) and fewer households living in poverty (NBELPOV100) will exhibit higher median house values. More recently, Leonard et al. (2016) apply hedonic price theory at the neighbourhood level and find that vacancy rates are negatively associated with property values, reflecting the role of housing abandonment and disinvestment as signals of neighbourhood decline — a dynamic particularly relevant in legacy cities such as Philadelphia. The proportion of single-family detached housing units (PCTSINGLES) is also included as a structural characteristic, as single-family homes typically command a premium over multi-family units in urban housing markets.

2. Methods

2a. Data Cleaning

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The original Philadelphia dataset contained 1,816 Census block groups. Prior to analysis, the dataset was cleaned by removing block groups that did not meet the following criteria: block groups with a total population of fewer than 40 residents, block groups with no housing units, and block groups where the median house value was lower than \$10,000. In addition, one outlier block group in North Philadelphia was removed, as it exhibited an unusually high median house value (exceeding \$800,000) alongside an exceptionally low median household income (below \$8,000), suggesting a data anomaly inconsistent with the broader sample. After cleaning, the final dataset used in all subsequent analyses contains 1,720 block groups.

2b. Exploratory Data Analysis

In this section, we examine the summary statistics and distributions of the dependent variable (median house value, MEDHVAL) and the four predictors: number of households living in poverty (NBELPOV100), proportion of residents with at least a bachelor's degree (PCTBACHMOR), proportion of vacant housing units (PCTVACANT), and proportion of detached single-family housing units (PCTSINGLES). Histograms of each variable are examined to assess distributional normality, and logarithmic transformations are applied where appropriate prior to regression analysis. We also examine scatter plots of the dependent variable against each predictor to assess linearity.

As part of the exploratory data analysis, we examine the pairwise correlations between the four predictors to assess potential multicollinearity. The Pearson sample correlation coefficient r measures the strength and direction of the linear relationship between two variables, and is calculated as:

$$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$$

The value of r ranges from -1 to $+1$, where values close to $+1$ indicate a strong positive linear relationship, values close to -1 indicate a strong negative linear relationship, and a value of 0 indicates no linear relationship between the two variables. Severe multicollinearity is flagged when $|r| > 0.8$ between any pair of predictors.

2c. Multiple Regression Analysis

Multiple linear regression is a statistical method used to model the relationship between a single continuous dependent variable and two or more independent predictor variables. The technique estimates the extent to which each predictor contributes to explaining variation in the dependent variable, while holding all other predictors constant. It is widely used in the social and spatial sciences to identify and quantify the associations between outcome variables and a set of explanatory factors.

In this analysis, the regression model takes the following form:

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$$\text{LNMEDHVAL} = \beta_0 + \beta_1(\text{PCTVACANT}) + \beta_2(\text{PCTSINGLES}) + \beta_3(\text{PCTBACHMOR}) + \beta_4(\text{LNNBELPOV100}) + \varepsilon$$

where β_0 is the intercept, representing the expected value of LNMEDHVAL when all predictors are equal to zero; β_1 through β_4 are the partial regression coefficients, each representing the expected change in LNMEDHVAL for a one-unit increase in the corresponding predictor, holding all other predictors constant; and ε is the error term, capturing the portion of variation in LNMEDHVAL that is not explained by the predictors.

The validity of OLS regression rests on several key assumptions. First, linearity assumes that the relationship between the dependent variable and each predictor is linear. Second, independence of observations assumes that the residuals are not correlated with one another. Third, normality of residuals assumes that the error terms are approximately normally distributed. Fourth, homoscedasticity assumes that the variance of the residuals is constant across all levels of the fitted values. Fifth, no multicollinearity assumes that the predictors are not too highly correlated with one another, which would otherwise make it difficult to isolate the individual effect of each predictor.

The parameters to be estimated in this model are β_0 , β_1 , β_2 , β_3 , β_4 , and σ^2 , where σ^2 denotes the variance of the error term ε , reflecting the overall variability of the observed values around the fitted regression line.

These parameters are estimated using the Ordinary Least Squares (OLS) method, which finds the values of β_0 , β_1 , ..., β_4 simultaneously to minimise the Error Sum of Squares (SSE), given by:

$$\text{SSE} = \sum_{i=1}^n \varepsilon^2 = \sum_{i=1}^n (y_i - \hat{y})^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \dots - \beta_k x_{ki})^2$$

where y_i is the observed value of LNMEDHVAL for block group i , $\hat{y}$ is the fitted value predicted by the model, and x_{1i} through x_{ki} are the values of the predictors for block group i . By minimising this expression, OLS produces the coefficient estimates that keep the total squared difference between observed and predicted values as small as possible.

For each predictor, the following hypotheses are tested:

$H_0: \beta_i = 0$ (the predictor has no effect on LNMEDHVAL)

$H_a: \beta_i \neq 0$ (the predictor has a significant effect on LNMEDHVAL)

A predictor is considered statistically significant if its associated p-value falls below the significance level of $\alpha = 0.05$, leading us to reject H_0 .

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The goodness of fit of the model is assessed using the coefficient of multiple determination R^2 , defined as:

$$R^2 = SSR / SST = 1 - (SSE / SST)$$

where SSR is the regression sum of squares representing variation in LNMEDHVAL explained by the model, SSE is the error sum of squares representing unexplained variation, and $SST = SSR + SSE$ is the total sum of squares. R^2 ranges from 0 to 1, with higher values indicating better model fit. Since R^2 increases with every additional predictor regardless of its contribution, the adjusted R^2 is also reported:

$$R^2_{adj} = 1 - [(SSE / (n - k - 1)) / (SST / (n - 1))]$$

where $n = 1,720$ is the number of observations and $k = 4$ is the number of predictors. The adjusted R^2 penalises the inclusion of predictors that do not meaningfully improve the model fit.

2d. Additional Analyses

To assess the predictive performance and generalisability of the regression model, 5-fold cross-validation is conducted. Cross-validation is a resampling technique used to evaluate how well a model performs on independent data, thereby guarding against overfitting — the tendency of a model to fit the training data too closely at the expense of predictive accuracy on new observations.

In 5-fold cross-validation, the dataset of 1,720 block groups is randomly partitioned into 5 equal subsets (folds). The model is trained on 4 of the 5 folds and tested on the remaining fold. This process is repeated 5 times, each time using a different fold as the test set, so that every observation is used for testing exactly once. The predictive errors from all 5 iterations are then combined to produce an overall measure of model performance.

Model performance is evaluated using the Root Mean Square Error (RMSE), which measures the average magnitude of the prediction errors. The RMSE is calculated as:

$$RMSE = \sqrt{[(1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2]}$$

where n is the number of observations, y_i is the observed value of LNMEDHVAL for block group i , and $\hat{y}_i$ is the corresponding predicted value. A lower RMSE indicates better predictive accuracy. The RMSE is used here to compare the performance of the full 4-predictor model against a reduced 2-predictor model (PCTVACANT and MEDHHINC), with the better-performing model indicated by the lower RMSE value.

2e. Software

All statistical analyses in this study are conducted using *R*. Choropleth maps of the dependent variable and predictors are created using *GeoDa*, an open-source spatial analysis software.

3. Results

3a. Exploratory Results

Table 1 presents the summary statistics for the dependent variable and four predictors. MEDHVAL has a mean of \$66,287.73 and a large standard deviation of \$60,006.08, reflecting considerable variation in house values across Philadelphia's block groups. Among the predictors, PCTBACHMOR averages 16.08% (SD = 17.77), PCTVACANT averages 11.29% (SD = 9.63), PCTSINGLES averages 9.23% (SD = 13.25), and NBELPOV100 averages 189.80 households (SD = 164.32).

Figure 1 presents histograms of the five untransformed variables. None of the variables appears approximately normally distributed, with all exhibiting notable skewness. **Figure 2** presents histograms of the log-transformed variables. The distribution of LNMEDHVAL appears approximately normal, exhibiting a roughly symmetric, bell-shaped curve centred around 11, indicating that the logarithmic transformation successfully normalises the dependent variable. LNMEDHVAL will therefore be used as the dependent variable in all subsequent regression analyses. For the predictors, the logarithmic transformation produces mixed results. LNNBELPOV100 is the only predictor that achieves an approximately normal distribution after transformation and will be used in place of NBELPOV100 in the regression. In contrast, LNPCBACHMORE, LNPCTVACANT, and LNPCTSINGLES all exhibit pronounced spikes at or near zero, indicative of zero-inflated distributions, and the original untransformed variables (PCTBACHMOR, PCTVACANT, and PCTSINGLES) will therefore be retained. The remaining regression assumptions will be examined in the Regression Assumption Checks section below.

Figure 3 presents scatter plots of LNMEDHVAL against each of the four predictors. The relationship between LNMEDHVAL and PCTVACANT appears negative, as block groups with higher vacancy rates tend to have lower median house values. However, the relationship does not follow a strictly linear pattern, with points heavily concentrated near zero and considerable spread at higher vacancy levels. Similarly, the relationship between LNMEDHVAL and PCTSINGLES shows a weak positive trend, though the distribution is heavily clustered near zero with no clear linear structure. The relationship between LNMEDHVAL and PCTBACHMOR is the most visibly linear of the four, showing a positive association whereby block groups with higher proportions of bachelor's degree holders tend to have higher median house values. Finally, LNMEDHVAL and LNNBELPOV100 exhibit a negative relationship, with the relationship appearing non-linear and flattening out at higher values. Overall, none of the four relationships appears strictly linear, which represents a potential violation of the linearity assumption and will be revisited in the Regression Assumption Checks section below.

Figure 4 presents the choropleth map of LNMEDHVAL across Philadelphia's 1,720 Census block groups. A clear spatial pattern is evident, with the highest median house values concentrated in the northwestern and northeastern areas of the city, while the lowest values are clustered in the north-central and central portions of Philadelphia.

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Figure 5 presents choropleth maps of the four predictors. The PCTVACANT map exhibits a spatial distribution that closely mirrors the inverse of LNMEDHVAL, with the highest vacancy rates concentrated in north-central Philadelphia where house values are lowest, suggesting a strong negative association. Similarly, the LNNBELPOV map shows high poverty concentrated in the same north-central areas, indicating a likely strong negative relationship with LNMEDHVAL. The PCTBACHMOR map shows higher educational attainment in the northwestern areas, broadly aligning with zones of higher house values and indicating a likely positive association. The PCTSINGLES map shows a somewhat different pattern, with higher proportions of single-family housing in the northeastern and outer areas of the city, showing a weaker visual correspondence with LNMEDHVAL. The spatial similarity between PCTVACANT and LNNBELPOV suggests these two predictors may be moderately correlated, though severe multicollinearity is not anticipated based on visual inspection alone.

Table 2 presents the Pearson correlation matrix of the four predictors. All pairwise correlations are modest in magnitude, ranging from -0.32 (PCTBACHMOR and LNNBELPOV100) to 0.25 (PCTVACANT and LNNBELPOV100), and no correlation exceeds $|r| = 0.8$. This confirms the absence of severe multicollinearity, consistent with the visual comparison of the predictor maps. The strongest negative correlation between PCTBACHMOR and LNNBELPOV100 ($r = -0.32$) aligns with the observation that the two maps show opposing spatial patterns. Given these results, all four predictors will be retained in the regression model.

Table 1. Summary Statistics of Variables

Variable	Mean	Std. Dev.
Median House Value (MEDHVAL)	66,287.73	60,006.08
# Households Living in Poverty (NBELPOV100)	189.80	164.32
% with Bachelor's or Higher (PCTBACHMOR)	16.08	17.77
% of Vacant Houses (PCTVACANT)	11.29	9.63
% of Single House Units (PCTSINGLES)	9.23	13.25

Table 2. Pearson Correlation Matrix of Predictors

	PCTVACANT	PCTSINGLES	PCTBACHMOR	LNNBELPOV100
PCTVACANT	1.000	-0.151	-0.298	0.250
PCTSINGLES	-0.151	1.000	0.198	-0.291
PCTBACHMOR	-0.298	0.198	1.000	-0.320
LNNBELPOV100	0.250	-0.291	-0.320	1.000

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Figure 1. Histograms of Variables (Untransformed)

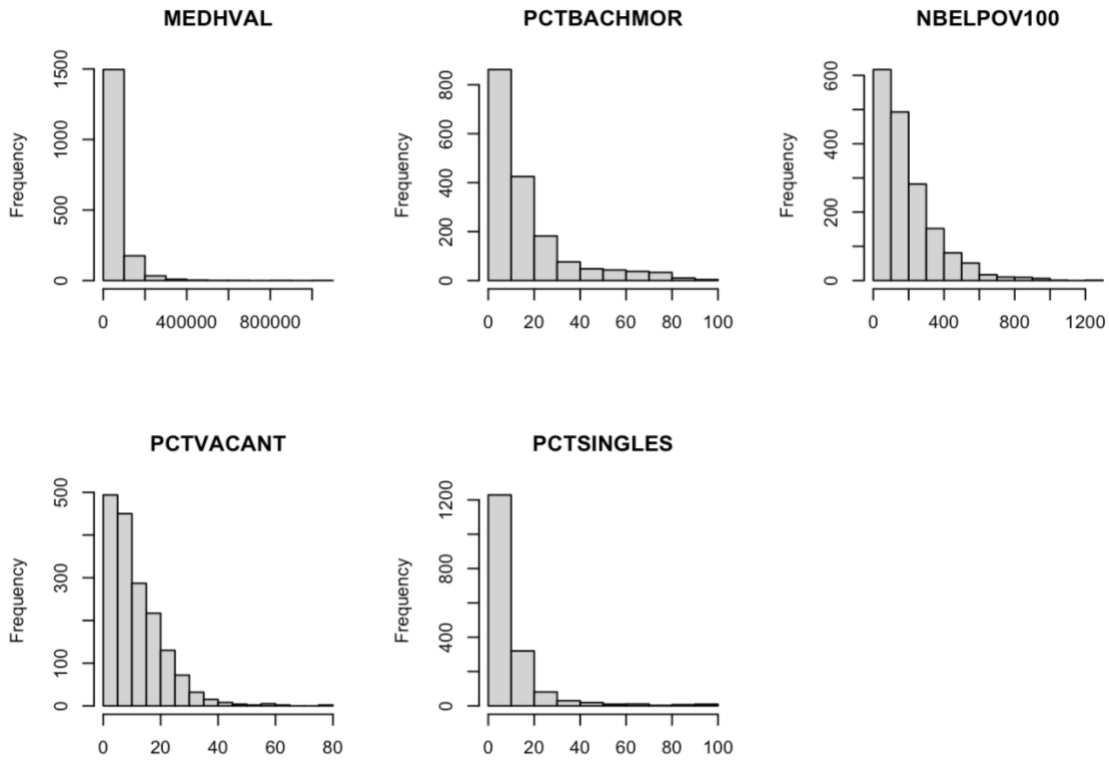
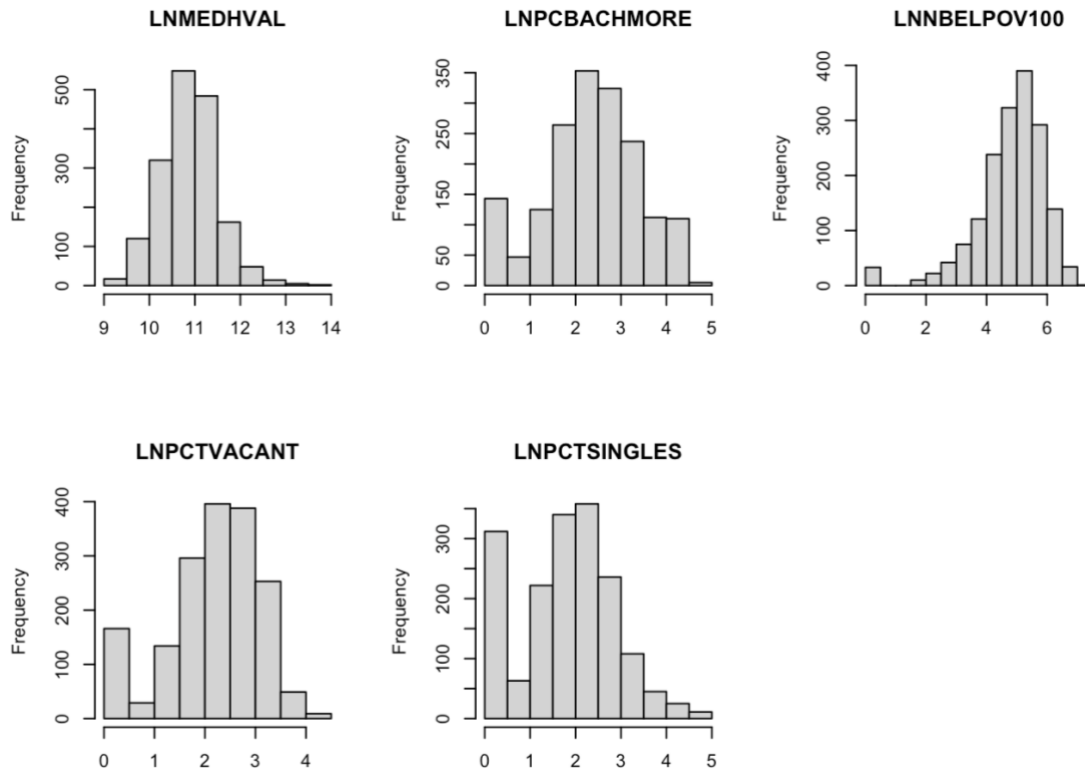


Figure 2. Histograms of Log-Transformed Variables



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Figure 3. Scatter Plots of Dependent Variable vs. Each Predictor

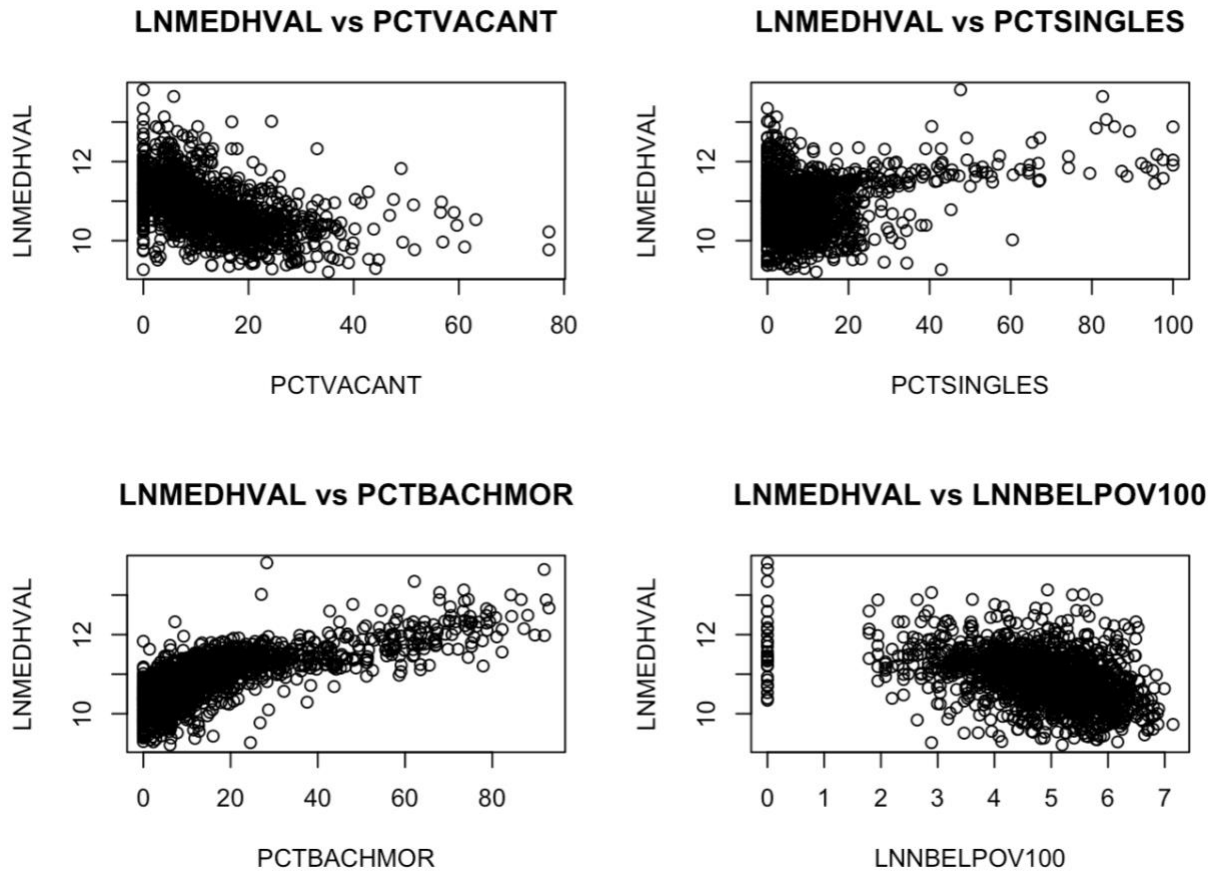


Figure 4. Choropleth Map of LNMEDHVAL

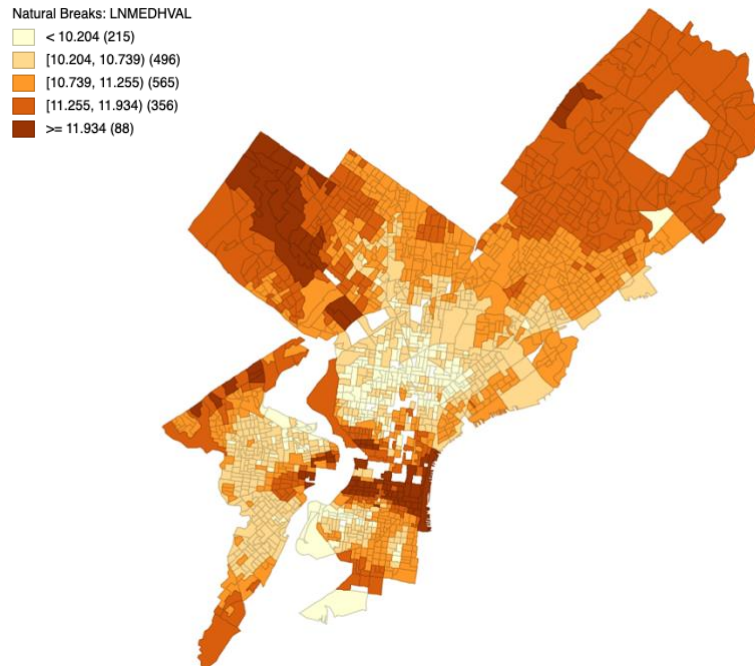


Figure 5. Choropleth Maps of Predictor Variables



3b. Regression Results

We regressed the log of median house value (LNMEDHVAL) on the proportion of vacant housing units (PCTVACANT), the proportion of detached single-family housing units (PCTSINGLES), the proportion of residents with at least a bachelor's degree (PCTBACHMOR), and the log of the number of households living in poverty (LNNBELPOV100). The regression output indicates that all four predictors are highly significant and are associated with median house values ($p < 0.001$ for all variables).

The regression output and ANOVA table are presented in **Figure 6**. The model estimates the following equation:

$$\text{LNMEDHVAL} = 11.114 - 0.019(\text{PCTVACANT}) + 0.003(\text{PCTSINGLES}) + 0.021(\text{PCTBACHMOR}) - 0.079(\text{LNNBELPOV100})$$

Since LNMEDHVAL is log-transformed, a one-unit increase in a predictor is associated with an approximate percentage change in median house value. A one percentage point increase in PCTVACANT is associated with a $\beta_1 = -0.0192$, or approximately a 1.92% decrease in median house value, holding all other predictors constant. A one percentage point increase in PCTSINGLES is associated with a $\beta_2 = 0.0030$, or approximately a 0.30% increase in median house value. A one percentage point increase in PCTBACHMOR is associated with a $\beta_3 = 0.0209$, or approximately a 2.09% increase in median house value. Since LNNBELPOV100 is also log-transformed, its coefficient is interpreted as an elasticity — a 1% increase in the number of households in poverty is associated with a $\beta_4 = -0.0789$, or approximately a 0.079% decrease in median house value.

The p-value of less than 0.001 for PCTVACANT tells us that if there is actually no relationship between PCTVACANT and LNMEDHVAL (i.e., if the null hypothesis that $\beta_1 = 0$ is actually true), then the probability of obtaining a β_1 coefficient estimate of -0.0192 is less than 0.001. The same logic applies to the remaining predictors where their p-values of less than 0.001 indicate that we can safely reject $H_0: \beta_2 = 0$, $H_0: \beta_3 = 0$, and $H_0: \beta_4 = 0$ in favour of $H_a: \beta_2 \neq 0$, $H_a: \beta_3 \neq 0$, and $H_a: \beta_4 \neq 0$, respectively (at most reasonable levels of α).

Over two-thirds of the variance in the dependent variable is explained by the model ($R^2 = 0.6623$, Adjusted $R^2 = 0.6615$). The low p-value associated with the F-ratio of 840.9 shows that we can reject the null hypothesis that all coefficients in the model are zero.

To further validate the model, stepwise regression was conducted using the AIC criterion, as presented in **Figure 7**. The stepwise procedure begins with the full 4-predictor model (AIC = -3448.07) and evaluates whether removing any predictor improves the AIC. Since removing any predictor results in a higher (worse) AIC, no variables are eliminated and the full model is confirmed as the best-fitting model. PCTBACHMOR contributes most to the model, as its removal would result in the largest AIC increase (from -3448.1 to -2379.0), followed by PCTVACANT

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(-3102.7), LNNBELPOV100 (-3364.9), and PCTSINGLES (s-3432.2). All four predictors are therefore retained in the final model.

Figure 6. Regression & ANOVA Output from R

```
Call:
lm(formula = LNMEDHVAL ~ PCTVACANT + PCTSINGLES + PCTBACHMOR +
    LNNBELPOV100, data = multregdata)

Residuals:
    Min       1Q   Median       3Q      Max
-2.25825 -0.20391  0.03822  0.21744  2.24347

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.1137661  0.0465330  238.836 < 2e-16 ***
PCTVACANT    -0.0191569  0.0009779  -19.590 < 2e-16 ***
PCTSINGLES    0.0029769  0.0007032   4.234 0.0000242 ***
PCTBACHMOR    0.0209098  0.0005432  38.494 < 2e-16 ***
LNNBELPOV100 -0.0789054  0.0084569  -9.330 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3665 on 1715 degrees of freedom
Multiple R-squared:  0.6623,    Adjusted R-squared:  0.6615
F-statistic: 840.9 on 4 and 1715 DF,  p-value: < 2.2e-16
```

```
Analysis of Variance Table

Response: LNMEDHVAL
      Df Sum Sq Mean Sq F value    Pr(>F)
PCTVACANT    1  180.392   180.392  1343.087 < 2e-16 ***
PCTSINGLES    1   24.543    24.543   182.734 < 2e-16 ***
PCTBACHMOR    1  235.118   235.118  1750.551 < 2e-16 ***
LNNBELPOV100    1   11.692    11.692    87.054 < 2e-16 ***
Residuals  1715   230.344     0.134
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 7. Stepwise Regression Output from R

```
Start: AIC=-3448.07
LNMEDHVAL ~ PCTVACANT + PCTSINGLES + PCTBACHMOR + LNNBELPOV100

      Df Sum of Sq  RSS   AIC
<none>      230.34 -3448.1
- PCTSINGLES    1    2.407 232.75 -3432.2
- LNNBELPOV100  1   11.692 242.04 -3364.9
- PCTVACANT     1   51.546 281.89 -3102.7
- PCTBACHMOR    1  199.020 429.36 -2379.0
```

```
> step_model$anova
  Step Df Deviance Resid. Df Resid. Dev      AIC
1     NA      NA      1715     230.3435 -3448.073
```

3c. Regression Assumption Checks

In this section, we examine the aptness of the regression model by testing the key OLS assumptions. The distributions of the variables were already examined in the Exploratory Results section above, where it was noted that logarithmic transformations were applied to MEDHVAL and NBELPOV100 to achieve more normal distributions prior to regression.

Figure 3 presents scatter plots of LNMEDHVAL against each of the four predictors. As discussed in the Exploratory Results section, none of the four relationships appears strictly linear. The relationships between LNMEDHVAL and PCTVACANT, and between LNMEDHVAL and LNNBELPOV100, are notably non-linear, with the spread of values fanning out at lower predictor values and flattening at higher values. The relationship between LNMEDHVAL and PCTSINGLES shows little discernible linear structure, while PCTBACHMOR exhibits the most linear trend of the four. Overall, the linearity assumption of OLS regression appears to be violated to some degree.

Figure 8 presents the histogram of the standardized residuals. Standardized residuals are the raw residuals divided by their standard deviation, allowing them to be interpreted on a common scale. The histogram shows a distribution that is approximately bell-shaped and centred near zero, suggesting that the normality of residuals assumption is reasonably satisfied. However, the distribution is slightly right-skewed with a notable spike at zero and some observations in the tails beyond ± 3 , indicating minor deviation from perfect normality.

Figure 9 presents the scatter plot of standardized residuals against predicted values. The points are broadly scattered around the horizontal zero line across the range of predicted values, which is consistent with the homoscedasticity assumption. However, the spread of residuals appears somewhat wider at lower predicted values (around 10) and narrows slightly at higher values, suggesting a mild degree of heteroscedasticity. Several outliers are visible with standardized residuals exceeding ± 3 , and a small number of extreme values reach beyond ± 5 , which may warrant further investigation. Nonetheless, given the large sample size of 1,720 block groups, the influence of these outliers on the overall model estimates is likely limited.

Finally, referencing the choropleth maps presented in **Figures 4 and 5** and the discussion in the Exploratory Results, there appears to be clear spatial patterning in both the dependent variable and the predictors — block groups with similar values tend to cluster together geographically. This suggests the presence of positive spatial autocorrelation, meaning that nearby block groups are more similar to one another than would be expected by chance. This represents a potential violation of the independence of observations assumption, as residuals in spatially proximate block groups may be correlated. Addressing spatial autocorrelation formally would require spatial regression techniques, which are beyond the scope of this analysis.

Figure 8. Histogram of Standardized Residuals

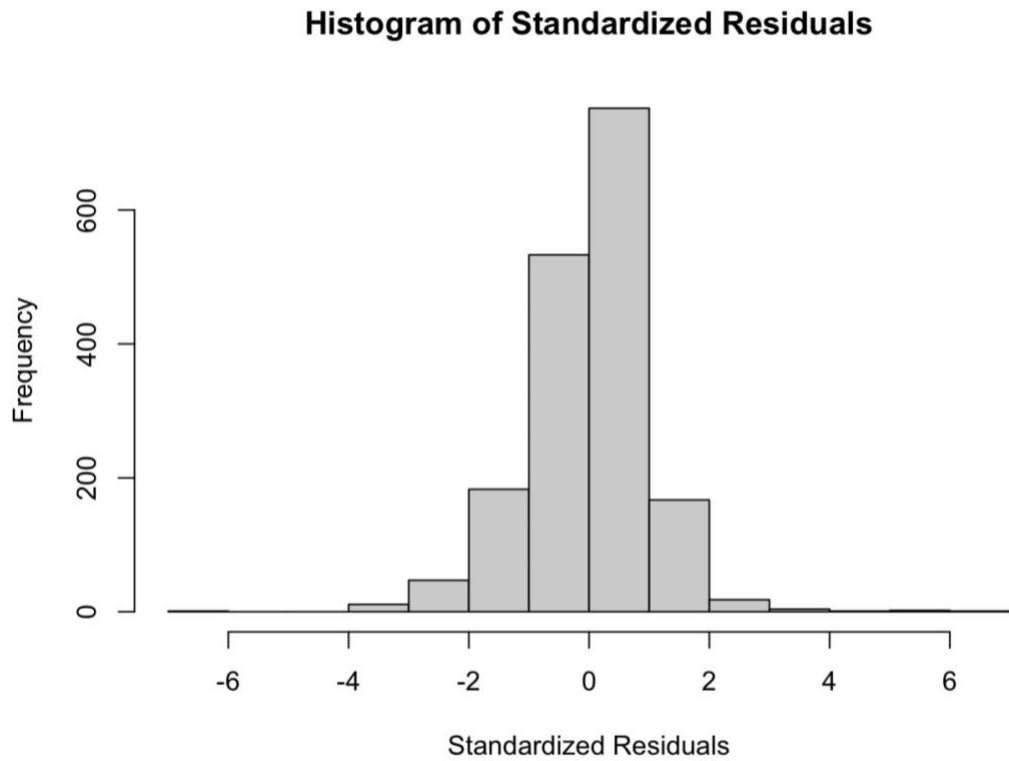
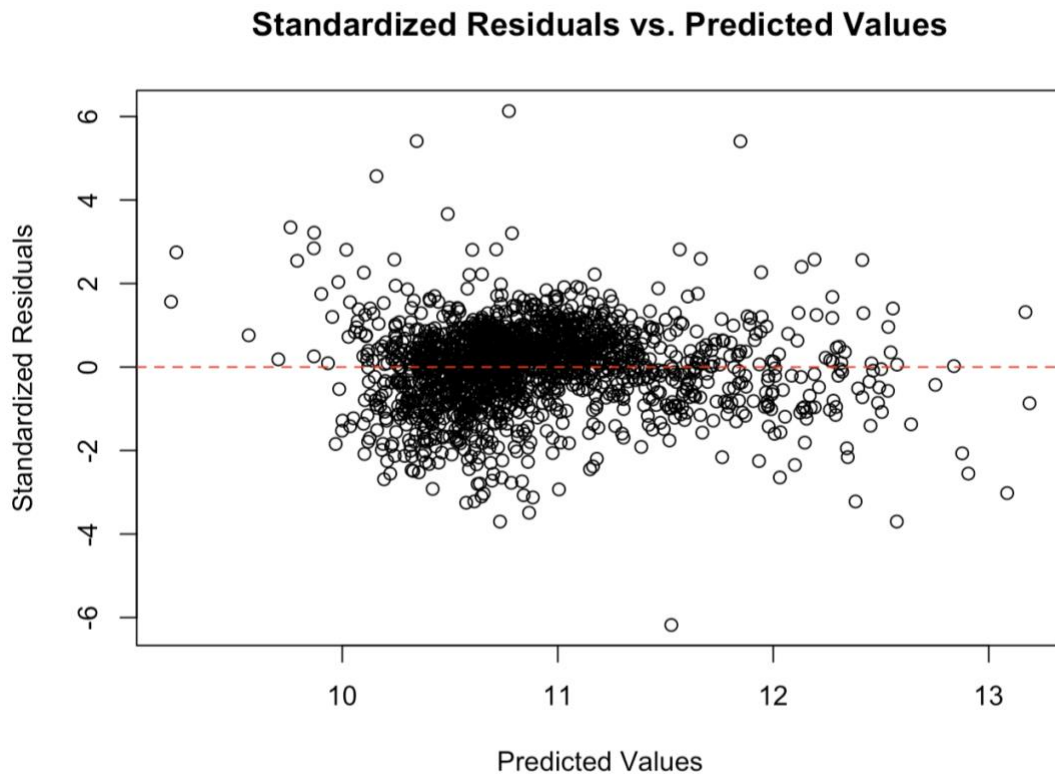


Figure 9. Standardized Residuals vs. Predicted Values



4. Discussion and Limitations

This study employed OLS multiple regression to model the log-transformed median house values (LNMEDHVAL) of census block groups in Philadelphia using four predictors: the percentage of vacant housing units (PCTVACANT), the percentage of single-family housing units (PCTSINGLES), the percentage of residents with a bachelor's degree or higher (PCTBACHMOR), and the log-transformed number of households below the poverty line (LNNBELPOV100). All four predictors were found to be statistically significant at the 0.001 level, confirming their role as meaningful determinants of residential property values.

The directions of the coefficients were largely consistent with theoretical expectations. PCTBACHMOR exerted the strongest positive effect on house values, reflecting the well-documented relationship between neighbourhood educational attainment and housing demand. PCTVACANT had the largest negative effect, as high vacancy rates signal disinvestment and reduce neighbourhood desirability. The negative coefficient on LNNBELPOV100 aligns with hedonic pricing theory where areas with greater concentrations of poverty tend to have lower property values; while the positive coefficient on PCTSINGLES was somewhat less intuitive, potentially reflecting the higher market value of single-family homes relative to denser housing typologies. It is assumed that these findings have implications for Philadelphia's housing policy, particularly in suggesting that reducing vacancy rates and addressing concentrated poverty in underserved neighbourhoods could help stabilize and improve residential property values.

The model demonstrated strong explanatory power, with an R^2 of **0.6623** and an adjusted R^2 of **0.6615**, indicating that **approximately two-thirds of the variance in log median house values is explained by the four predictors**. The overall F-statistic of 840.9 ($p < 2.2e-16$) confirms the joint significance of the model. Figure 10 presents a stepwise regression using AIC as the selection criterion that suggests retaining all four predictors in the final model, it can be interpreted that none of the variables were redundant given the others. Five-fold cross-validation further supported the superiority of the full four-predictor model (RMSE = 0.3664) over a reduced two-predictor model comprising only PCTVACANT and MEDHHINC (RMSE = 0.4427), indicating better out-of-sample predictive accuracy. Additional variables that could potentially improve the model include proximity to transit, crime rates, school quality ratings, and distance to the central business district, all of which are theoretically linked to residential property values but were unavailable in the current dataset.

Several limitations should be acknowledged. First, the assumption of linearity was not fully satisfied for all predictors, as evidenced by the scatter plots of PCTVACANT and LNNBELPOV100 against LNMEDHVAL in **Figure 3**, which showed some non-linear patterns. Second, the residual map in **Figure 9** suggests the presence of positive spatial autocorrelation in the residuals, which, if confirmed, would violate the OLS assumption of independence. This suggests that spatially proximate block groups share unmeasured characteristics that the model fails to capture, and a spatial regression model may be more appropriate. Third, there was mild evidence of heteroscedasticity, with the variance of residuals appearing somewhat smaller at lower

fitted values, which can lead to inefficient coefficient estimates and unreliable standard errors. Future work should address these violations through various spatial modelling techniques.

Figure 10. Cross-Validation Results (CVlm, k=5)

```
# 4-predictor model
cv_full <- CVlm(data = multregdata,
               form.lm = formula(LNMEDHVAL ~ PCTVACANT + PCTSINGLES +
                                PCTBACHMOR + LNNBELPOV100),
               m = 5, printit = FALSE)

rmse_full <- sqrt(mean((cv_full$LNMEDHVAL - cv_full$cvpred)^2))
rmse_full
[1] 0.3664401

# 2-predictor model
cv_reduced <- CVlm(data = multregdata,
                  form.lm = formula(LNMEDHVAL ~ PCTVACANT + MEDHHINC),
                  m = 5, printit = FALSE)

rmse_reduced <- sqrt(mean((cv_reduced$LNMEDHVAL - cv_reduced$cvpred)^2))
rmse_reduced
[1] 0.4427216
```

5. References

Dubin, R.A. and Goodman, A.C. (1982). *Valuation of education and crime neighborhood characteristics through hedonic housing prices*. Population and Environment, 5(3), 166–181.

Gyourko, J., & Summers, A. A. (2007). *Residential land use regulation in the Philadelphia MSA* (Working Paper No. 560). Zell/Lurie Real Estate Center, Wharton School, University of Pennsylvania.

Leonard, T., Powell-Wiley, T.M., Ayers, C. and Murdoch, J.C. (2016). *Property values as a measure of neighborhoods*. American Journal of Preventive Medicine, 50(4), 494–503.

Li, M.M. and Brown, H.J. (1980). *Micro-neighborhood externalities and hedonic housing prices*. Land Economics, 56(2), 125–141.